

Studio 2

Introduction to Object Oriented Programming

Objectives

To look at class design and how to create simple classes. The student should be able to identify properties and methods for a class and how to implement them.

Exercise 1

1. Talk to your neighbour. Discuss your understanding of the following terms.
 - a) Class
 - b) Object
 - c) Attribute/Property/Field
 - d) Method
 - e) Constructor
2. What is the difference between the private and public keywords when used in a class? Which one is better to use?
3. Does the programmer need to create a constructor for every class?
4. Nilesh needs an application to store details on his extensive Blu-Ray movie collection. Each Blu-Ray movie has a title, genre (Action, Thriller, Drama, Science Fiction), year released and who it has been lent to (blank if not lent out).
 - a) Create a Bluray class and then create private fields to store the above information.
 - b) Create the appropriate constructor for the class passing everything to the constructor except the lent to value. This should be set to an empty string.
 - c) Create a toString method to return all information back joined up as one string with a space between each value.
 - d) Create get and set methods for the lent to field.

5. Write the code to create BluRay objects for the following two awesome movies:

Serenity, Science Fiction, 2005
Despicable Me, Comedy, 2010

Now write the code to display the information from the objects to test that they have been created correctly.

Now write the code to change the object for Serenity so that it stores that it has been lent out to Tim, re-display the objects information. Now display the name Serenity and who it has been lent to.

6. Create a class called Triangle which stores the 3 corners of a triangle using PVector objects and also stores the line colour for the triangle.

The class should have a constructor which is passed the x and y coordinates of all the corners and the line colour.

It should also have a toString method to get all the coordinate information about the triangle.

It will also need a method which will draw the triangle in the sketch window in the correct colour.

7. In the setup method create a triangle object and draw it to the sketch window.
8. Create a variable to store a collection of triangle objects at class scope level. Then add 3 triangles to the collection in the setup method.
9. Write the code to display the info about each triangle to the console window.
10. Write the code to draw each triangle in the sketch window.

Exercise 2

1. A university stores the following information about all students enrolled at the university:

Id (int), name, address, paid fees (boolean).

Using the sketch from Ex 1, add a new class called Student and create private fields for the above information.
2. Create a constructor for the class which is passed the id, name and address and uses those values to initialise the object. The paid fees field should be set to false.
3. Create a get methods for all fields and also a set method for the paid fees field so that it can be changed if required.
4. Modify the get method for the paid fees field so that it returns back "Has Paid" if they have paid their fees or "Has not paid" if they haven't paid their fees.
5. Create a toString method which returns back all the information joined up as one string with a space between each value. Instead of joining the word True or False to the string it should display the appropriate message as described in Ex 4. How can you easily do this?
6. Create 3 student objects in the setup method and make up some information. Display all the information about the first student to the console window. Now set the Paid fees field to true for the first student and re-display the information to the console window.
7. Create a collection object to store all the student objects and then add the students to the collection. Write the code in the setup method that will display all information about all students in the collection to the console window.
8. Extend the setup method to display just the name and if they have paid their fees or not for each student in the collection to the console window.

Exercise 3

1. Create a new processing sketch and in the setup method set the size of the window to something reasonably large.
 2. Create a new class called button which has private fields to store the x and y position of the centre of the button as a PVector object and fields to store the width and height of the button and to store the colour of the button.
- Hint: Don't use the variables names 'width' and 'height' as they are built in variables and will show up pink.
3. Create the constructor which is passed the x and y position, the width and height and also the colour to initialise the object.
 4. Create a draw method that will draw the button to the sketch window.
- Hint: Set the rect mode to CENTER before drawing.
5. In the sketch code create a collection to store button objects at class scope level and in the setup method create and add 3 buttons to the collection.
 6. In the draw method, draw all the buttons in the collection to the sketch window.
 7. Test your sketch and make sure all buttons are drawn.

Exercise 4 (Studio Verification Exercise)

1. Create a new processing sketch and in the setup method set the size of the window to something reasonably large.
2. Create a new class called Lollipop. The class needs to store the x and y position of the middle of the lollipop using a PVector object, the colour of the lollipop and the colour of the handle. Create constants for the size of the lollipop (width and height) and set it to 60, the width of the handle set to 4 and height of the handle set to 60.
3. Create a constructor which is passed the x and y position of the lollipop and uses the values to initialise the PVector object. The constructor should set the colour of the lollipop to a random colour. Investigate the color method on the Processing website to help with this. Set the colour of the handle to a whiteish colour.

4. Create a method to draw the lollipop. It should draw the handle with the x and y position being the top centre of the handle. Then draw the lollipop centred around it's x and y position.
5. Create a new lollipop object in the setup method and draw it to the sketch window to test your code, then comment the test code out.
6. Create a collection to store lollipop objects at class scope level.
7. Investigate the built in mousePressed method on the Processing website. Create the mousePressed image in your sketch code and write the code so that whenever the left mouse button is pressed a new lollipop is added to the collection.
8. Create the draw method in the sketch code so that all of the lollipops in the collection are drawn to the sketch window.
9. Test your code and make sure new lollipops are added whenever you click the left mouse button.